

Week 3: Introduction to Data Analysis - Working with Real Data

1. Install pandas library using simple pip command
2. Download a simple dataset (like student marks or sales data)
3. Load the dataset and display first few rows
4. Check basic information: how many rows, columns, data types
5. Find and handle missing values in the dataset
6. Calculate average, highest, lowest values for numerical columns

Project:

Analyze a Simple Sales Dataset to find: total sales, best-selling product, and create a basic report

Tips & Resources:

- Search YouTube: 'Pandas for absolute beginners'
- Use small datasets first (less than 100 rows)
- Practice with data from your daily life (expenses, study hours)
- Don't get overwhelmed - focus on one concept at a time
- Use ChatGPT to explain errors in simple language

1. Install pandas library using simple pip command

```
In [1]: pip install pandas
```

```
Requirement already satisfied: pandas in c:\users\lenovo\appdata\local\programs\python\python311\lib\site-packages (2.3.1)
Requirement already satisfied: numpy>=1.23.2 in c:\users\lenovo\appdata\local\programs\python\python311\lib\site-packages (from pandas) (1.26.4)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\lenovo\appdata\local\programs\python\python311\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\lenovo\appdata\local\programs\python\python311\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\lenovo\appdata\local\programs\python\python311\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\lenovo\appdata\local\programs\python\python311\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

2. Download a simple dataset (like student marks or sales data)

```
In [2]: ## Downloaded Amazon Store Sales Data
import numpy as np
import pandas as pd
```

3. Load the dataset and display first few rows

```
In [3]: # Loading the dataset

data = pd.read_csv("Amazon Store Sales Data.csv")

print("----- THE FIRST 5 RECORDS OF DATA -----")
print(data.head(5))
```

```

----- THE FIRST 5 RECORDS OF DATA -----
  Row ID+O6G3A1:R6      Order ID Order Date Ship Date      Ship Mode \
0      4918  CA-2019-160304  1-1-2019  1-7-2019  Standard Class
1      4919  CA-2019-160304  1-2-2019  1-7-2019  Standard Class
2      4920  CA-2019-160304  1-2-2019  1-7-2019  Standard Class
3      3074  CA-2019-125206  1-3-2019  1-5-2019    First Class
4      8604  US-2019-116365  1-3-2019  1-8-2019  Standard Class

  Customer ID      Customer Name      Segment      Country      City ... \
0  BM-11575      Brendan Murry  Corporate  United States  Gaithersburg ...
1  BM-11575      Brendan Murry  Corporate  United States  Gaithersburg ...
2  BM-11575      Brendan Murry  Corporate  United States  Gaithersburg ...
3  LR-16915      Lena Radford   Consumer  United States  Los Angeles ...
4  CA-12310      Christine Abelman  Corporate  United States  San Antonio ...

  Region      Product ID      Category Sub-Category \
0  East  FUR-BO-10004709      Furniture  Bookcases
1  East  FUR-BO-10004709      Furniture  Bookcases
2  East  TEC-PH-10000455      Technology  Phones
3  West  OFF-ST-10003692  Office Supplies  Storage
4  Central  TEC-AC-10002217      Technology  Accessories

      Product Name      Sales  Quantity \
0  Bush Westfield Collection Bookcases, Medium Ch...  73.94      1
1  Bush Westfield Collection Bookcases, Medium Ch... 173.94      3
2      GE 30522EE2  231.98      2
3  Recycled Steel Personal File for Hanging File ... 114.46      2
4      Imation Clip USB flash drive - 8 GB  30.08      2

  Profit  Returns  Payment Mode
0  28.2668      0      Online
1  38.2668      0      Online
2  67.2742      0      Cards
3  28.6150      0      Online
4  -5.2640      0      Online

```

[5 rows x 21 columns]

4. Check basic information: how many rows, columns, data types

```

In [4]: print("\n ----- Dataset Shape (Rows, Columns):-----")
        print(data.shape)

        print("\n----- Column Names: -----")
        print(data.columns)

        print("\n----- Data Types: -----")
        print(data.dtypes)

        print("\n----- Full Info:-----")
        data.info()

```

----- Dataset Shape (Rows, Columns):-----
(5901, 21)

----- Column Names: -----
Index(['Row ID+06G3A1:R6', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode',
 'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State',
 'Region', 'Product ID', 'Category', 'Sub-Category', 'Product Name',
 'Sales', 'Quantity', 'Profit', 'Returns', 'Payment Mode'],
 dtype='object')

----- Data Types: -----
Row ID+06G3A1:R6 int64
Order ID object
Order Date object
Ship Date object
Ship Mode object
Customer ID object
Customer Name object
Segment object
Country object
City object
State object
Region object
Product ID object
Category object
Sub-Category object
Product Name object
Sales float64
Quantity int64
Profit float64
Returns int64
Payment Mode object
dtype: object

----- Full Info:-----
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5901 entries, 0 to 5900
Data columns (total 21 columns):
Column Non-Null Count Dtype
--- ---
0 Row ID+06G3A1:R6 5901 non-null int64
1 Order ID 5901 non-null object
2 Order Date 5901 non-null object
3 Ship Date 5901 non-null object
4 Ship Mode 5901 non-null object
5 Customer ID 5901 non-null object
6 Customer Name 5901 non-null object
7 Segment 5901 non-null object
8 Country 5901 non-null object
9 City 5901 non-null object
10 State 5901 non-null object
11 Region 5901 non-null object
12 Product ID 5901 non-null object
13 Category 5901 non-null object
14 Sub-Category 5901 non-null object
15 Product Name 5901 non-null object
16 Sales 5901 non-null float64
17 Quantity 5901 non-null int64
18 Profit 5901 non-null float64
19 Returns 5901 non-null int64
20 Payment Mode 5901 non-null object
dtypes: float64(2), int64(3), object(16)
memory usage: 968.3+ KB

5. Find and handle missing values in the dataset

```
In [5]: print("\n -----Missing Values in Each Column:-----")
        print(data.isnull().sum())
```

```
-----Missing Values in Each Column:-----
Row ID+06G3A1:R6      0
Order ID              0
Order Date            0
Ship Date             0
Ship Mode             0
Customer ID           0
Customer Name         0
Segment              0
Country              0
City                 0
State                0
Region               0
Product ID           0
Category             0
Sub-Category         0
Product Name         0
Sales                0
Quantity             0
Profit               0
Returns              0
Payment Mode         0
dtype: int64
```

6. Calculate average, highest, lowest values for numerical columns

```
In [6]: print("\n ----- Summary Statistics for Numerical Columns: -----")
        print(data.describe())

        # Manually calculate:
        print("\n----- Averages: -----")
        print(data.mean(numeric_only=True))

        print("\n----- Maximum values: -----")
        print(data.max(numeric_only=True))

        print("\n----- Minimum values:-----")
        print(data.min(numeric_only=True))
```

```

----- Summary Statistics for Numerical Columns: -----
      Row ID+06G3A1:R6      Sales      Quantity      Profit      Returns
count      5901.000000  5901.000000  5901.000000  5901.000000  5901.000000
mean      5022.422471   265.345589    3.781901    29.700408    0.048636
std      2877.977184   474.260645    2.212917   259.589138    0.215124
min         1.000000     0.836000    1.000000  -6599.978000    0.000000
25%      2486.000000    71.976000    2.000000    1.795500    0.000000
50%      5091.000000   128.648000    3.000000    8.502500    0.000000
75%      7456.000000   265.170000    5.000000   28.615000    0.000000
max      9994.000000  9099.930000   14.000000  8399.976000    1.000000

```

```

----- Averages: -----
Row ID+06G3A1:R6      5022.422471
Sales                  265.345589
Quantity               3.781901
Profit                 29.700408
Returns                0.048636
dtype: float64

```

```

----- Maximum values: -----
Row ID+06G3A1:R6      9994.000
Sales                  9099.930
Quantity               14.000
Profit                 8399.976
Returns                1.000
dtype: float64

```

```

----- Minimum values:-----
Row ID+06G3A1:R6      1.000
Sales                  0.836
Quantity               1.000
Profit                -6599.978
Returns                0.000
dtype: float64

```